

Leverage Large Language Models For Enhanced Aviation Safety

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Abstract—The modernization of the National Airspace System presents numerous opportunities for increasing the capacity for safe aircraft navigation through the exploitation of AI/ML capabilities. In 2021, the authors outlined the promise and challenges of Artificial Intelligence (AI) technology applications in The Journal of Air Traffic Control cover article, “Novel Framework to Advance Verification, Validation, and Certification of Non-Deterministic, AI-Based Algorithms for Safety Critical Applications.”^[1] Since 2021, the field of AI has been transformed by new breakthroughs in AI/ML technology, such as the surge in Large Language Model (LLM) technology within the field of natural language processing. This intersection of LLM technology and natural language processing affords an automated ability to interpret, analyze, translate, and even generate language.

This Paper looks at this technology-application intersection in the context of air traffic control. An AI/ML-enabled aircraft/ATC alerting system, combined with enhanced positional information, can verbalize timely notifications about impending unsafe situations allowing aircraft flight crews to take proactive measures. ATC systems can also benefit from AI/ML-enabled safety monitoring to ensure all parties are informed about real-time events.

TABLE I. LLM ENHANCED SAFETY ALERTING SYSTEM DEMONSTRATION PLAN

Step	Objective
1	Analyze the language-based air-ground communication that transacts between the ground-based air traffic control (ATC) function and flight crews operating aircraft;
2	Compare the ‘predicted’ flight operation authorized by ATC against the ‘observed’ (surveilled) operation performed by the flight crews;
3	Assess the operational conformance of observed flight operations against regulatory safety requirements, e.g., Airspace Classifications, Flight Standards, etc.; and
4	Provide safety alerting, in real-time, to ATC and aircraft encountering compromised safety margins.

This paper presents one promising approach to explore Step 1 (see Table 1) that employs deep learning and LLMs to create a system that can process air-ground verbal transactions, detect anomalous situations, and alert air traffic controllers, thus enhancing ATC and flight crew situational awareness based on identified anomalies raised in air traffic communications. Our results indicate that cost-effective

data can be generated with LLMs and then used to train a text-based Variational Auto-Encoder that can successfully discriminate between nominal and off-nominal (safety-critical) scenarios observed in air traffic communication.

Keywords— *Safety assurance, verification, validation of non-deterministic behavior*

I. INTRODUCTION

Aviation safety is under constant pressure to improve, while being challenged by airspace complexity; high-density flight operations; technology refresh and infrastructure modernization; proximal operation of legacy aircraft with new entrants; and continued pressure to improve operating margins while navigating increasingly narrow safety margins. An area where recent data^[2] indicates at best correlation, and potentially causation, is safety margin impairment due to human factors.

Recent reports (e.g., the FAA-commissioned Safety Review Team report^[2]) have drawn attention to the disturbing frequency of documented near-misses occurring on a near-daily basis within the NAS. Whether it’s a garbled radio message, loss of situational awareness, insufficient training, or sensory overload; our NAS is under serious threat of a catastrophic safety escape.

On February 4, 2023, a runway incursion and overflight occurred at the Austin-Bergstrom International Airport (KAUS) when a FedEx flight and Southwest Airlines flight failed to maintain separation. (See Fig. 1 from the National Transportation Safety Board (NTSB) preliminary report on the incident.^[3]) This incident involved a descending aircraft (FDX1432/767-32LF) that received a clearance to land at KAUS RWY 18L. Shortly thereafter, another aircraft (SWA708/737-79P) received a clearance to taxi onto the same active runway (KAUS RWY 18L) and depart. Although safety investigators have yet to issue their findings, publicly available surveillance data indicate these two aircraft, having both received operational clearances from ATC to use KAUS RWY 18L at roughly the same time, were operating well below separational minimums.

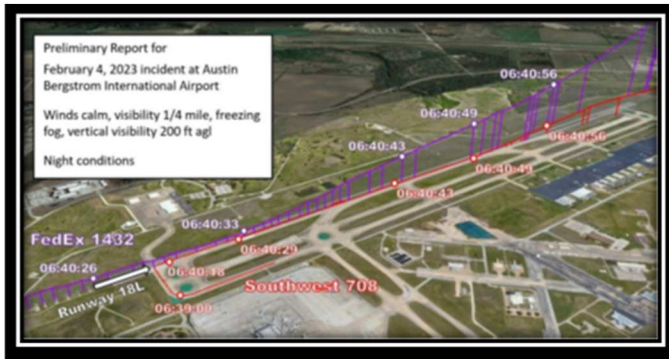


Fig. 1. Conflicted arrival at KAUS (from the NTSB preliminary report [3])

While each safety event cited is unique, clear patterns of causation exist. One nexus of safety-escape causation is the disconnect that can occur between NAS procedures, situational awareness, and low-fidelity air-ground communications. This nexus is further exacerbated when differing levels of operational proficiency interact or procedural deviations by either a flight crew or air traffic controller occur. Most of these occur with negligible safety margin impairment. However, like any systems analysis, eroding safety margins can create a quality or operational escape under certain circumstances. Under this situation, it would be helpful to have some executive oversight to draw attention to a margin degradation and need for corrective action.

That confluence of system safety escapes and their coupling to human factors has led us to investigate the applicability of Large Language Models (LLMs) and Deep Learning in Aviation System Safety. Specifically, this paper examines the potential for air-ground communications analyses enabled by the coupling of text analytics trained on synthetic data generated by LLMs with real-time surveillance, NAS procedural constraints, and airspace management objectives. Fig. 2 shows our proposed use case for using AI/ML to proactively identify safety risks from air traffic communications and then alert ATCs and flight crews to the need for mitigating action.

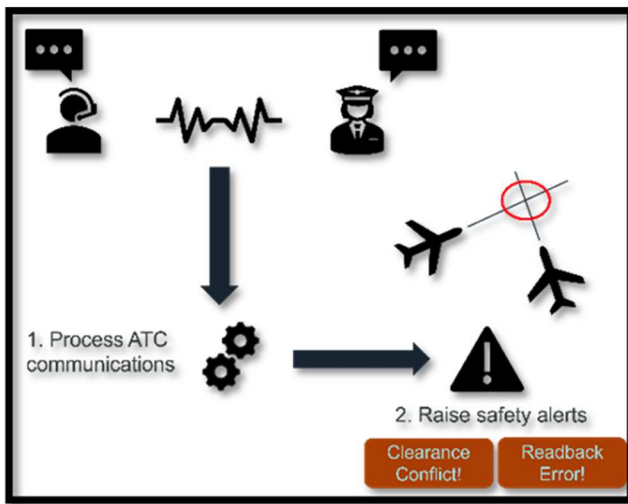


Fig. 2. Use case for monitoring air traffic communications for safety risks

II. LARGE LANGUAGE MODELS BACKGROUND

A. Overview

Language Models (LMs) are representations of natural human language and are typically used to predict the next word in a sequence [4] [5] (see Fig. 3). Large Language Models (LLMs) are enabled by the Transformer model architecture [6], which is a set of neural networks (NNs) consisting of an encoder and a decoder with self-attention capabilities. The encoder and decoder extract meanings from a sequence of text and understand the relationships between words and phrases in the text. Transformer LLMs perform self-learning, and through this process learn to understand basic grammar, languages, and knowledge [7]. Transformers process entire sequences in parallel, allowing data scientists to use GPUs to train transformer-based LLMs, reducing training time of the final language model. The Transformer NN architecture enables the use of very large language models with billions of parameters. The resulting LLMs can be used to process and respond to complex human-like prompts by generating a human-like response in a natural human language [8]. LLMs have facilitated the general trend of increasing performance on natural language tasks by increasing the number of parameters trained by the language model [4].

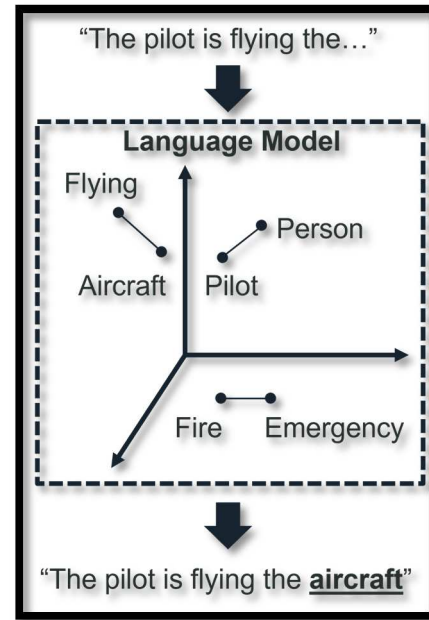


Fig. 3. Natural language generation with a Language Model

LLMs were recently popularized by OpenAI's public release of ChatGPT (Chat Generative Pre-trained Transformer) in late 2022, surpassing 100 million active users within two months of its release [9]. Any task which can be expressed in natural language can be addressed to some degree by an LLM-enabled system. Such tasks, as identified in [8] [10], include:

1. content creation
2. semantic understanding
3. natural language inference
4. text classification
5. sentiment analysis
6. dialogue

7. question answering
8. language translation
9. text summarization
10. software development and debugging
11. personalized content recommendations

B. Training and Emergent Behaviors

There are two approaches for training LLMs for a given task: traditional fine-tuning and prompting^[11]. In traditional fine-tuning, an LLM is first trained on a large corpus of unlabeled text data through a process called *pretraining* and then it is further trained to be knowledgeable about a specific domain through supervised learning on a domain-specific corpus through a process called *fine-tuning*. In prompt-based training, a pretrained LLM is adapted to a specific task or domain through zero-shot or few-shot learning. Few-shot learning provides a few examples of desired input-output examples or additional information through preliminary prompts to the LLM before requesting responses from the LLM in subsequent prompts^[12]. In contrast, zero-shot learning or augmented prompting^[12] employs strategies to obtain specific outputs by carefully framing the prompt without providing examples a priori. An example of augmented prompting is chain of thought (CoT) prompting, which prompts the LLM to provide a chain of statements explaining intermediate reasoning that leads the LLM to its ultimate conclusion^{[11][13]}. Fig. 4 shows the process of obtaining a pretrained LLM and from it obtaining either a fine-tuned LLM through traditional fine-tuning or an assistant model through few-shot learning.

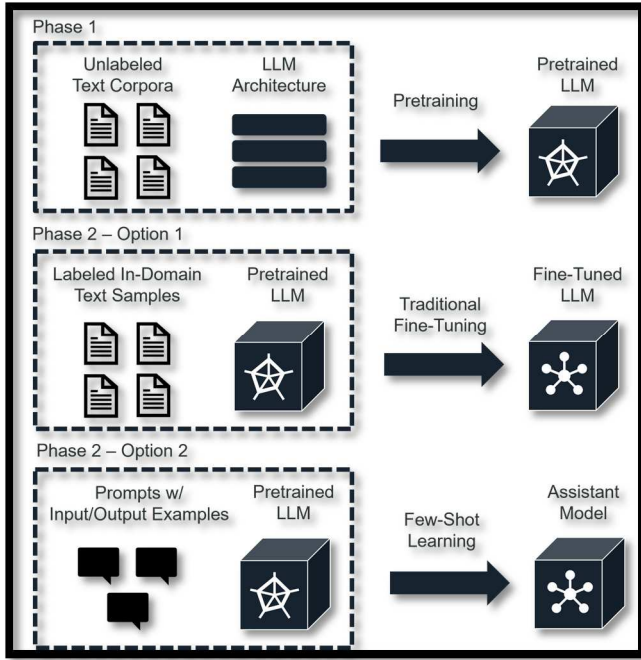


Fig. 4. Training a Large Language Model

Few-shot and zero-shot learning are considered emergent behaviors of LLMs, which are capabilities that manifest as a result of training a sufficiently large model rather than explicitly designing or programming the model to possess or learn such capabilities^[12]. Other emergent abilities that appear in LLMs of

sufficient size include multi-task language understanding, instruction following in new tasks, and program execution^[12].

C. Explaining LLM Behavior

With the rapid increase in capabilities of LLM-enabled systems comes the increased responsibility to ensure ethical use of these systems, verification of their outputs, and a proper understanding of their strengths and limitations.

The exploration of procedures that provide explanations of specific outputs and overall implicit reasoning capabilities in traditionally fine-tuned LLMs is an ongoing area of active research, which can be divided into two groups: local explainability and global explainability^[11]. Local explainability seeks to understand what led an LLM to produce a given output for a specific prompt. Global explainability seeks to understand the overall reasoning process that is employed by the LLM. Table II contains the taxonomy of LLM explainability techniques from^[11].

TABLE II. EXPLAINABILITY TECHNIQUES FOR LARGE LANGUAGE MODELS

TYPE	TECHNIQUE	OBJECTIVE
LOCAL EXPLAINABILITY	Feature Attribution-Based	Measure the impact of individual words on final output.
	Attention-Based	Identify the most relevant inputs to the final output.
	Example-Based	Demonstrate how alterations to individual inputs affect the output.
	Natural Language	Natural language explains decision-making process for a given output.
GLOBAL EXPLAINABILITY	Probing-Based	Understand which linguistic patterns are learned by different layers of the LM.
	Neuron Activation	Understand which neurons in the LM account for different types of outputs.
	Concept-Based	Identify how the presence of specific concepts impacts model predictions.
	Mechanistic Interpretation	Assess how connections between neurons result in predictions.

Furthermore, the *goodness* of a local or global explanation is measured in terms of *plausibility* to a human and *faithfulness* to the LLM's actual reasoning process^[11].

D. LLM Limitations and Vulnerabilities

Active research is also being done to understand the limitations of LLMs. The first limitation involves the possibility of LLMs generating *hallucinations* or confident assertions of factually incorrect or unverifiable information by the LLM^[14]. Hallucinations remain a concern because users who assume LLM-enabled systems will provide accurate information or predictions may make adverse decisions with safety critical implications based on such information. Hallucinations arise from limitations in the LLM model architecture or in deficiencies in the dataset used to train the LLM^[14]. Data set deficiencies include lacking relevant data about the domain or repeating data, which causes the LLM to memorize data rather than learn to reason. LLMs can be made more robust against generating hallucinations by including information from an external source in the prompt or by using chain-of-thought prompts^[13]. LLMs may also fall victim to sycophancy, the

predilection of generating outputs that align with a user's views rather than outputs that are factually correct^[15], and LLMs may propagate stereotype biases found in their training corpora^[8].

LLMs are also susceptible to adversarial attacks or attempts to use them for unethical purposes. Two such attacks are jailbreaking and data extraction attacks. Jailbreaking is the act of circumventing mechanisms the LLM employs to avoid generating undesirable content, such as instructions for how to commit a crime^[16]. One such jailbreaking approach, Greedy Coordinate Gradient (GCG), achieves this by combining prompt engineering (i.e., asking the LLM to begin its response with, "Sure! Here's the answer...") with a programmatically determined suffix to jailbreak LLMs, even proprietary ones, with any adversarial prompt^[16]. (A defense to this attack, SmoothLLM, has since been proposed in^[17].)

LLMs are also prone to memorizing certain portions of their training data, making it possible for attackers to extract sensitive or personal identifiable information, such as a person's phone number^[18]. Larger LLMs are especially vulnerable to a data extraction attack, even if the sensitive information was only present in a single training document. Practices proposed in^[18] that can mitigate the risk of LLMs falling victim to the data extraction attack include selecting sources for training data that do not contain sensitive information, fine-tuning an LLM to "forget" memorized data, explicitly auditing LLMs for memorization, and employing differential privacy techniques^[19] at the expense of longer training times and decreased model accuracy^[18].

E. Summary

LLMs have achieved impressive capabilities in recent years and have captured public attention and industrial investment. Their high performance in several natural language tasks has opened several opportunities for exploiting the capabilities of LLMs in many domains, including aviation safety. New emergent capabilities that arise from the development of better LLMs will continue to make this technology more effective and accessible. Active research is seeking to understand how LLMs learn to reason while also understanding their limitations and how to shore up their vulnerabilities, thus promoting safe and ethical use and paving the way for establishing industry-wide verification and validation standards for LLMs. We thus explore the incorporation of this technology into aviation safety systems to determine the art of the possible.

III. VAE BACKGROUND

Variational Auto Encoders (VAEs) are a useful machine learning method for anomaly detection. We combine text analytics with a VAE and analyze the latent space to determine outliers in an aviation safety application. A VAE assumes that the source data has some sort of underlying probability distribution (such as Gaussian) and then attempts to find the parameters of the distribution^[20].

A VAE network architecture is comprised of two parts, an encoder and a decoder. The encoder is responsible for learning a mapping from raw text data to a low-dimensional latent-space encoding that explains as much of the data as possible. The decoder is responsible for learning the inverse mapping that takes as input a single sample from the latent space and

reconstructs the original text input. Fig. 5 shows the components of a VAE. VAEs learn a probability distribution on the latent space such that all text inputs in a training dataset presented to the VAE can be encoded into the learned latent space and then successfully decoded back into the original text.

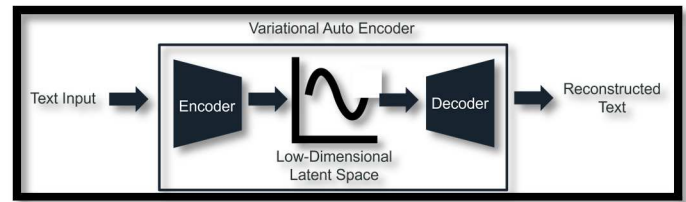


Fig. 5. Reconstructing Text Data with a Variational Auto Encoder

Optimizing latent space using VAEs provides a way to help analysts test anomaly detection hypotheses. Anomaly detection strategies seek to measure reconstruction loss, which measures how "different" from a typical "normal" sample a given text input appears to the VAE. During training, the VAE learns from text samples labeled as "normal". When unlabeled samples are presented to the VAE during testing, the model uses its encoder to convert the text samples into their latent space encoding, and then it decodes them back to text. The "reconstruction loss" represents the difference between the original text and the reconstructed text. Normal text samples will have low reconstruction loss. Anomalous text samples cannot be reconstructed sufficiently, and reconstruction loss increases^[21].

IV. PROPOSED FRAMEWORK

Our proposed framework (see Fig. 6) seeks to enhance the situational awareness capabilities of air traffic controllers by monitoring real-time ATC communications, identifying anomalous and potentially safety-critical situations, and raising an appropriate alert to the cognizant air traffic controller and flight crews. In this way, our system will act as a "wingman" to controllers to confirm or elevate missed risk potentials, decreasing time to decision and providing a more effective focus for controllers. With the addition of LLMs, safety alerts may also contain recommended actions for mitigating identified risks.

The act of processing human natural language in air traffic communication requires specific inputs and processing steps to determine whether a given situation requires increased attention by an ATC. We thus propose the Aviation Safety Anomaly Detection framework in Fig. 6. Here, we have two input frameworks: the Air Traffic Communication Framework and the Operational Regulatory Framework. The Air Traffic Communication Framework converts audio streams from air-ground communication into analyzable text transcriptions, which can then be processed by LLMs in an LLM virtual space to dynamically identify relevant actors, entities, speaker intentions, and other information pertinent to evaluating a situation's safety status. The Operational Regulatory Framework is a static codification of the aviation operational standards that govern industry expectations and dictate flight safety constraints. These are defined in joint orders and other command media enforced by the FAA. Both frameworks are available to the anomaly detection pipeline that monitors communications for discrepancies in expected operational behavior.

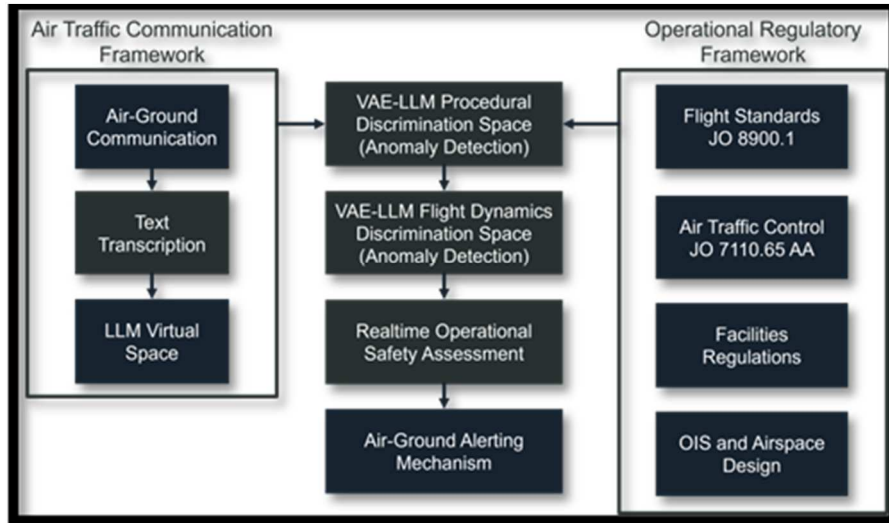


Fig. 6. Proposed Aviation Safety Anomaly Detection Framework

To determine whether a situation requires increased attention or intervening action by a controller or flight crew, two questions must be answered:

1. Are all aircraft and flight crews compliant with orders issued by the air traffic controller?
2. Does the execution of ATC orders enable all aircraft to maintain safe separation and operations for the duration of operational execution?

In the anomaly detection pipeline, two modules answer both of these questions in the procedural discrimination space and flight dynamics discrimination space, respectively, before performing a real-time operational safety assessment. During this assessment, if a safety concern has been identified, then an appropriate alerting mechanism is activated to notify the appropriate air traffic controller and flight crews that the immediate situation requires their attention, and that action should be taken to mitigate safety risks. The nature of the alert (e.g., text-based display, light display, auditory signal) will depend on the role and personal preferences of all actors involved. In this paper, we describe a novel text-based anomaly detection implementation for air traffic communications that forms the heart of this safety framework.

V. TEXT ANALYTICS ANOMALY DETECTION

In this section, we outline the text analytics anomaly detection pipeline that we implemented to discriminate between nominal and off-nominal air traffic communications. The version of the pipeline presented here is a standalone anomaly detection module corresponding to the VAE-LLM Procedural Discrimination Space module shown earlier in Fig. 6. This version of the module processes conversations received from the Air Traffic Communication Framework; future work will allow the module to also reference the contents of the Operational Regulatory Framework.

A. Text Preprocessing

Our anomaly detection framework combines both text analytics and anomaly detection methodologies. Fig. 7 shows our anomaly detection training and evaluation flow diagram. We first preprocess incoming text data to add parts of speech, normalize words, erase punctuation, and remove stop words. Next, we convert the words to numeric encodings to allow the data to be more easily processed by the VAE (Step 1 of Fig. 7). Then, a two-dimensional representation (size 30x30) of the numerical encodings is generated (Step 2) and then up-sampled to a larger image (size 128x128, Step 3) to take advantage of convolutional filters, which are used in our VAE model architecture.

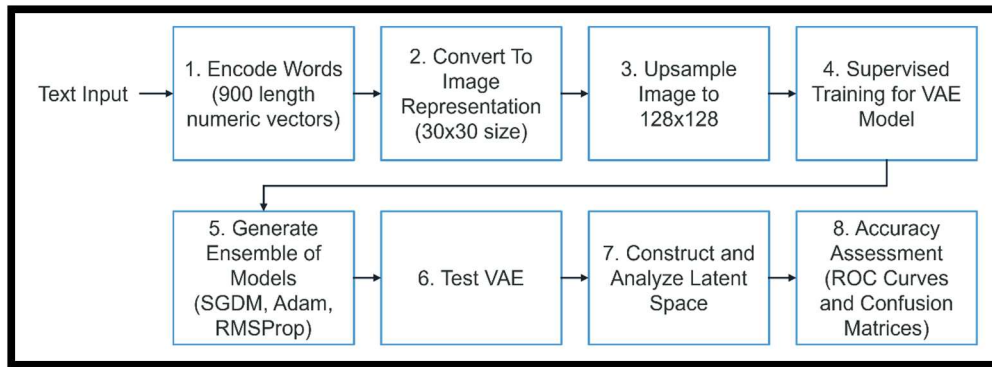


Fig. 7. Text Analytics Anomaly Detection – Training and Evaluation Flow Diagram

B. VAE Model Architecture

We use supervised learning (Step 4) to generate a VAE model with the characteristics of nominal air traffic communications. Once the VAE model has been trained, it will be able to process air traffic communications and determine if they corresponds to a nominal or off-nominal scenario.

Fig. 8 shows the deep learning layers used in the encoder and decoder components of the VAE model architecture. The VAE

encoder generates a compressed representation of the input data utilizing various weights and biases. Weights are the parameters within a neural network that transform input data within the network's hidden layers. A neural network is made up of a series of nodes. Within each node is a set of inputs, weights, and bias values. The weights and bias values in the nodes of each layer are modified during machine learning.

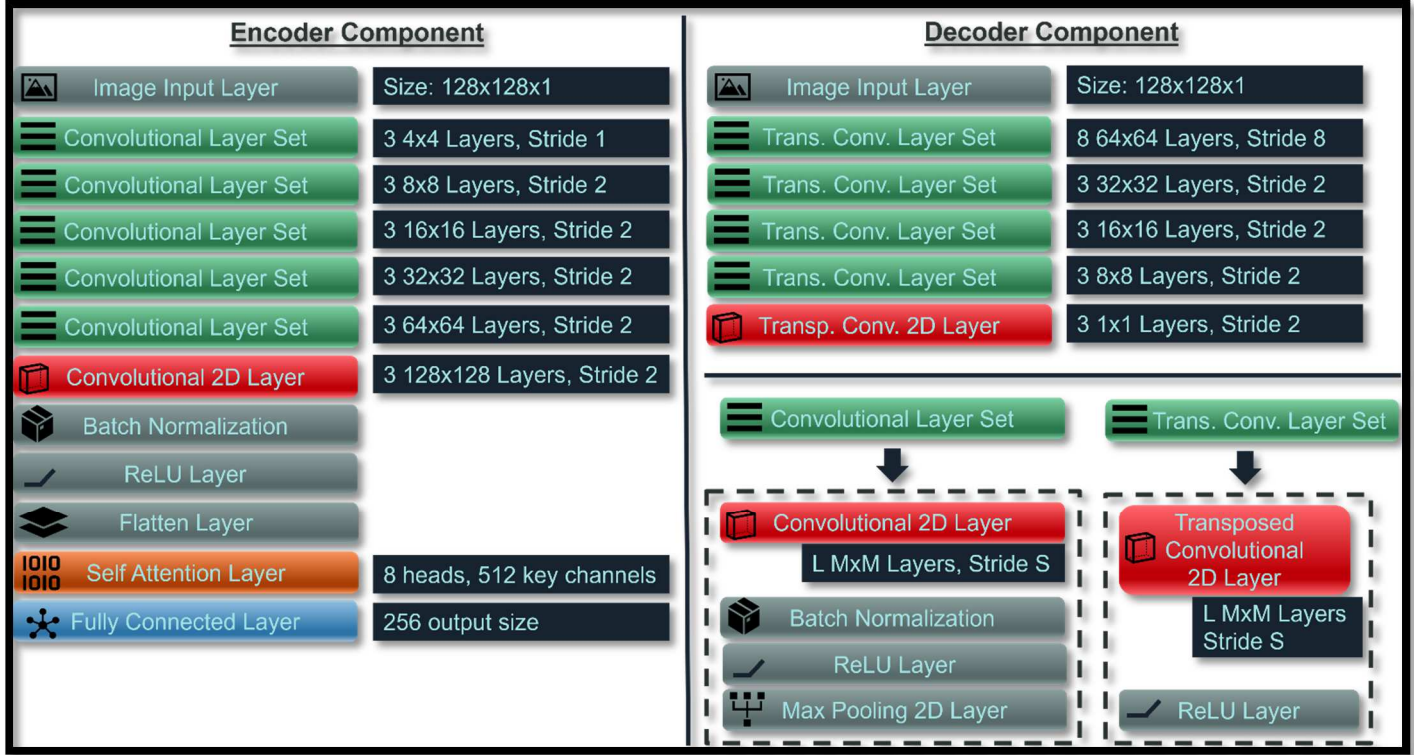


Fig. 8. VAE Model Architecture

One beneficial layer in the VAE models is multi-head self-attention layer [6]. The heads help encode contextual relationships between types of words, such as between adjectives and nouns, between adverbs and verbs, between bigrams, etc. This layer enhances the accuracy performance with deep connections in the layers. We observed that it requires various amounts of increased memory and processing times. We used eight attention heads, 512 channels, and 128 dimensions as model parameters.

The compressed representation of the input data is called the hidden vector. The mean and variance from the hidden vector are sampled and learned by the convolutional neural network (CNN), which processes images encoding the text data. The convolutional layers are helpful in determining the similarity of different features.

C. Ensemble Learning

We next create an ensemble of models with different solvers (Step 5) with each model using the VAE model architecture. We used three different gradient descent solvers: Adam, Stochastic Gradient Descent with Momentum (SGDM), and Root Mean

Square Propagation (RMSprop). Fig. 9 shows the creation of our three different models using the VAE architecture.

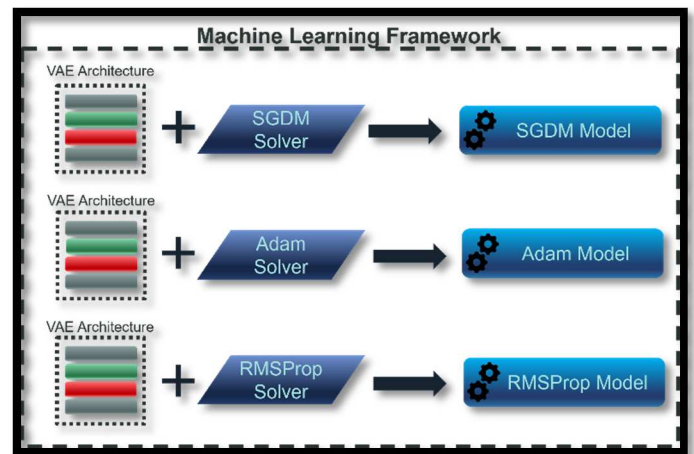


Fig. 9. Machine Learning Framework for Model Generation

We train each model with 80% of the normal test samples and test with 20% of the normal samples plus the abnormal samples. The VAE only needs to train with normal data. A given text sample is determined by the VAE to be normal or anomalous based on how different it is compared with a typical normal sample.

An optimal game theoretic linear program implementation can then be used to choose the solver from an ensemble of models with least error. The best model to believe for prediction can be chosen on a per sample basis. We use a linear program to determine which model to weight more heavily. This optimization can perform better than any single classifier. Linear optimization is useful for solving game theory problems and finding optimal strategies.

A reward matrix ‘A’ is constructed and solved with linear programming. The A matrix that we use consists of confidence or error values from predicted responses of several ML classifiers. Once the best classifier has been chosen, it is applied to the test data. This ensures that the best algorithm is always used to process incoming data. We use an interior-point algorithm, the primal-dual method, which must be feasible for convergence [22].

Given the normal/anomalous predictions of all three VAE models for a given text sample, a single most promising prediction from the ensemble of solvers can be generated, as exemplified in Fig. 10.

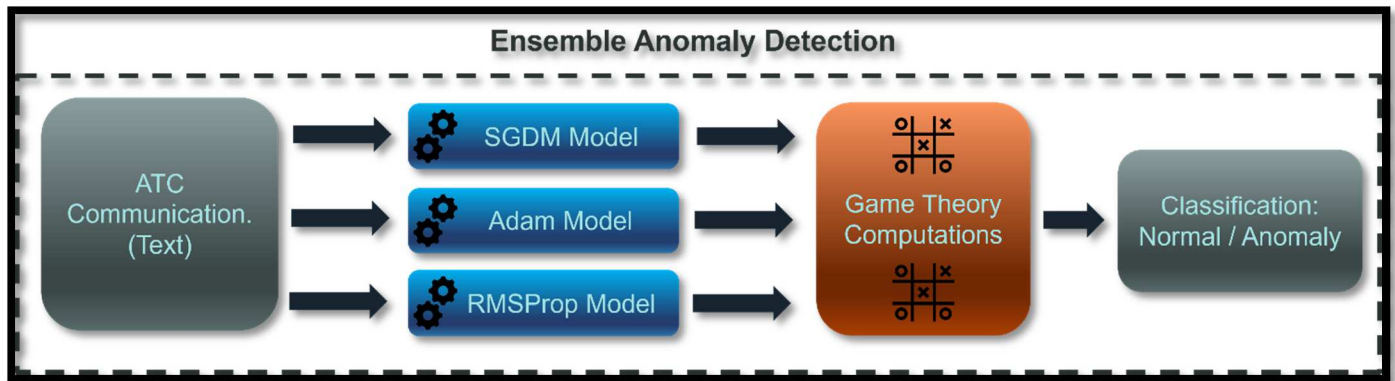


Fig. 10. Ensemble Anomaly Detection

D. Evaluation

Given the ensemble anomaly detection system, we then measure how well our system can distinguish between normal and anomalous text samples. This is done by plotting the samples in our dataset within a visualized latent space and then analyzing the proportion of normal and anomalous text samples that our system correctly classifies.

We use the statistical z-test and analyze the p-value to determine if a given test sample belongs to the normal data distribution (Step 6). A sample is determined to be an anomaly if the likelihood probability of its membership in the normal sample distribution is sufficiently small. We use significance levels (p-values) to determine if the new sample belongs to the normal distribution. In null hypothesis significance testing, the p-value is the probability of obtaining test results at least as extreme as the results observed, under the assumption that the null hypothesis (i.e., that the text sample is normal) is correct. A very small p-value means that such an extreme observed outcome would be very unlikely under the null hypothesis.

Principal component analysis (PCA) of the hidden vector allows for the visualization of n-dimensional point clusters, preferably 3-D point clusters, in the latent space in order to observe and compute maximum separation (Step 7). If the VAE learned successfully, then it will be able to plot all data samples within its internal latent space such that all normal data samples appear close together while anomalous data points are further away from the normal samples cluster. Fig. 11 shows two example visualizations – one where a VAE has successfully

learned an internal latent space representation that successfully separates anomalous data samples from normal ones, and one where the VAE has learned poorly, and normal samples are not separable from anomalous ones in the learned latent space.

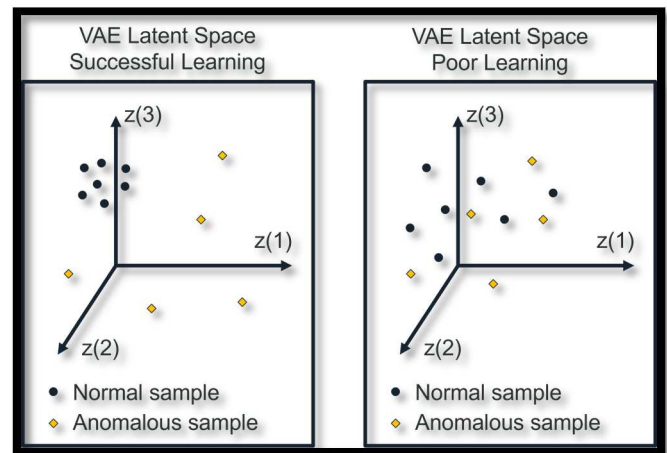


Fig. 11. VAE Learning-Quality Convolution in Latent Space

To determine the system's overall accuracy across the entire test set, we compute Receiver Operating Characteristic (ROC) curves. ROC curves are widely accepted as the standard way to judge performance. Given various combinations of normal and anomalous samples, the ROC curve measures how well the system correctly classifies normal samples as normal while

correctly classifying anomalous samples as anomalous. The Area Under the Curve (AUC) ^[23] is used to provide a single summary performance metric for the ROC curve. The closer it is to 1, the better the prediction. The AUC is commonly used to compare the performance of different classifiers.

VI. DATA

A. Overview

This section describes the datasets we considered for our research and how we preprocessed our selected dataset for use by our anomaly detection algorithm. For our work, we wanted data that captured *normal* and *anomalous* scenarios, the latter involving events that indicate a safety risk or required intervention by an air traffic controller. We analyzed three datasets:

1. ATCO public dataset ^[24]
2. LiveATC dataset ^[25]
3. LLM-generated synthetic dataset

B. Terminology

For this discussion, we consider a *conversation* to be the collection of time-adjacent utterances by at least two speakers. A conversation has a single arbitrary starting and ending point. A single *utterance* is audio communication (one or more sentences) by exactly one speaker that begins and ends in silence. Consecutive utterances can be made by the same speaker if there is silence between utterances. While it is possible that two or more speakers may be speaking simultaneously, we assume for simplicity that such occurrences are rare.

For our work, we classify a single conversation as *normal* or *nominal* if it does not contain verbal indicators of elevated safety risks or abnormal aircraft operational behavior. A conversation is considered *anomalous* or *off-nominal* if a safety risk is identified by one of the speakers or if a situation requires further inquiry or guidance from the air traffic controller.

C. ATCO Dataset

The ATCO dataset ^[24] was curated by transcribing 1.1 hours of audio communication between English-speaking pilots and air traffic controllers. The dataset is composed of multiple independent conversations, which we informally observed as having fewer than 10 utterances per conversation. This dataset was created to support research in Automatic Speech Recognition (ASR). We ultimately decided that this dataset did not suit the needs of our work because it lacked clear anomalous examples, and the conversations were too short for meaningful use by our algorithm.

D. LiveATC Dataset

LiveATC ^[25] is an archive of air traffic audio communications collected by volunteers and made available for public use. The audio is livestreamed, but interesting recordings of anomalous situations are also collected in the archive for historical purposes. We acquired audio samples from the LiveATC archive, transcribed them with TurboScribe ^[26], and then manually edited the transcriptions to include information about who was speaking for each utterance. However, the

transcription and manual editing process proved too labor-intensive for collecting a sufficiently large dataset.

E. LLM-Generated Synthetic Dataset

For our synthetic dataset, we used the OpenAI API to auto-generate sample conversations between an air traffic controller and one or more pilots in both normal and anomalous situations. The gpt-3.5-turbo model was used with a temperature of 0.90 to increase the diversity of sample conversations. (A temperature of 0.0 has no diversity while a temperature of 1.0 has maximum diversity.)

For normal situations, we provided ChatGPT with this prompt: “Give me a sample conversation between an air traffic controller and one or more pilots in a random situation, using fake callsigns, fake runway numbers, etc. Use at least 2000 words. Let ATC be Air Traffic Controller and let P be Pilot. Only output dialogue.”

For anomalous situations, we provided these prompts:

1. Give me a sample conversation between an air traffic controller and one or more pilots.
2. Use fake callsigns, fake runway numbers, etc.
3. Use at least 2000 words.
4. Let ATC be Air Traffic Controller and let P be Pilot.
5. Only output dialogue.
6. Include a word count at the end; and
7. In the middle of the conversation, have <anomalous situation> occur. (The <anomalous situation> component of the prompt had one of 11 values (see Table III).)

TABLE III. ANOMOLOUS SITUATION DIALOGUE EXEMPLAR

Ref#	Anomalous Situation
1	“... the pilot notices that something is on fire.”
2	“... there is a bird strike.”
3	“... the pilot notices a pothole on the runway.”
4	“... the pilot notifies the controller that there are pressurization issues.”
5	“... the pilot has to make an emergency landing.”
6	“... the pilot reads the controller's directions back incorrectly.”
7	“... the pilot accidentally deviates from the controller's instructions.”
8	“... there is sudden engine failure in the aircraft.”
9	“... the pilot raises a concern about extreme weather.”
10	“... a pilot makes a declared emergency, requiring the controller to adjust the flight paths of one or more aircraft.”
11	“... two aircraft come too close together.”

We generated a dataset with 550 normal conversations and 550 anomalous conversations (evenly divided across the 11 anomalous situation types). Each conversation was converted into a single string with a delimiter separating utterances and another delimiter separating a speaker from the message content. This is an example conversation: “ATC :: Delta Niner Niner, you are cleared to land. ;; Pilot :: Cleared to land, Delta Niner Niner. ;; ...”

Table IV includes partial snippets of conversations that were generated by ChatGPT for each anomalous scenario. The ability to auto-generate a sufficiently large dataset with diverse and humanlike conversational scenarios led us to use this LLM-generated synthetic dataset for our aviation safety application.

TABLE IV. EXAMPLES - SYNTHETIC (CHATGPT) AIR-GROUND COMMUNICATION - ANOMALOUS SITUATION

Ref ID	Anomalous Situation	Partial Conversation Snippet	
		Speaker	Message
1	Fire	Pilot	Roger, Departure Control. Alpha Bravo Charlie will maintain heading 270 and climb to FL200.
		Pilot	ATC, this is Alpha Bravo Charlie. I have a cockpit warning indicator showing a possible fire in the cargo compartment. Requesting immediate return to the airport.
		ATC	Alpha Bravo Charlie, this is ATC. Roger. Turn right heading 090 and descend to 3000 feet. I will coordinate for an emergency landing. Standby for further instructions.
2	Bird Strike	ATC	Bluebird 752, this is ATC. Be advised, we have reports of a large flock of birds in your vicinity, please maintain a lookout.
		Pilot 1	Copy that, ATC. We will maintain a lookout for the birds, Bluebird 752.
		ATC	Skyhawk 234, be advised, we have received reports of a bird strike in the vicinity of your departure. Please use caution.
		Pilot 2	Roger that ATC. We will be vigilant for any birds, Skyhawk 234.
		Pilot 1	ATC, this is Bluebird 752, we've had a bird strike! We've lost an engine and need to make an emergency landing!
3	Runway Pothole	Pilot 1	Climbing to 7,500 feet and maintaining present heading, Skyhawk 375.
		Pilot 1	Austin Departure, Skyhawk 375. We have a visual on a pothole at the midpoint of Runway 17L. Can you confirm?
		ATC	Skyhawk 375, this is Austin Tower. We will have the maintenance crew check the runway immediately. Thank you for the report.
4	Aircraft Pressurization Issue	ATC	Air France 789, roger, I'll advise when you can proceed.
		Pilot 1	Roger, holding position, Air France 789.
		Pilot 2	Heathrow Tower, Speedbird 456, we have a pressurization issue and need to descend immediately.
		ATC	Speedbird 456, understood. Descend and maintain 5,000 feet, turn left heading 270 degrees.
5	Emergency Landing	Pilot	ATC, Flight 123 here. We have a mayday situation. Engine failure. Requesting immediate emergency landing instructions.
		ATC	Flight 123, roger. I see you on radar. You are about 20 miles east of our airport. Maintain heading 270 and descend to 4,000 feet. You are cleared for an emergency landing on Runway 09L. Wind is calm.
6	Pilot Readback Error	ATC	Good morning, Atlas 123, this is Tower. You are cleared for takeoff on Runway 27R. Wind is calm, altimeter is 29.92. Departure frequency is 118.1. Have a safe flight.
		Pilot	Tower, Atlas 123, roger. Cleared for takeoff on Runway 27L. Departure frequency 118.1. Thank you.
		ATC	Correction, Atlas 123, you are cleared for takeoff on Runway 27R, not 27L. Confirm you copy.
		Pilot	Tower, Atlas 123, my apologies. Copy that. Cleared for takeoff on Runway 27R. Departure frequency 118.1. Thank you.
7	Pilot Deviation	ATC	ALPHA-ONE, wind 180 degrees at 15 knots. Runway 16L is now available for landing. Continue holding for another 5 minutes, then expect vectors for the ILS approach runway 16L, over.
		Pilot	Holding for another 5 minutes, then expecting vectors for the ISL approach runway 16L. Thank you, Los Angeles Center. ALPHA-ONE.
		ATC	ALPHA-ONE, it appears you've deviated from the holding pattern. Correct immediately and maintain the assigned hold, over.
		Pilot	Apologies, Los Angeles Center. We had a slight misunderstanding. We will correct our position and maintain the assigned hold, ALPHA-ONE.
8	Engine Failure	Pilot	Atlanta Approach, Delta Bravo Charlie. Mayday! Mayday! We are experiencing engine failure. Requesting immediate assistance. Over.
		ATC	Delta Bravo Charlie, Atlanta Approach. Roger. Stay calm. Maintain your current heading. We will provide vectors for an emergency landing. Over.
9	Extreme Weather	Pilot	Departure, Flight 123. We are currently encountering some turbulence. Requesting a deviation to the north if possible.
		ATC	Flight 123, roger that. You are cleared to deviate to the north as requested. Report when you are back on course.
		Pilot	Cleared to deviate to the north. We will report when back on course. Thank you, Departure.
10	Declared Emergency	Pilot 5	ATC, this is Tango 789, requesting immediate landing clearance due to an emergency.
		ATC	Tango 789, understood. Squawk 7700. Cleared to land runway 22L. Emergency services will be standing by.
		Pilot 5	Squawking 7700. Cleared to land runway 22L. Tango 789.
		ATC	American 789, cancel takeoff clearance. Hold position on runway 13L.
		Pilot 3	Holding position on runway 13L. American 789.
		ATC	United 456, go around. Climb and maintain 4,000 feet. Turn right heading 300.
11	Aircraft Too Close	Pilot 2	Going around. Climbing and maintaining 4,000 feet. Turning right heading 300. United 456.
		ATC	Alpha Bravo Charlie, this is Air Traffic Control. Traffic alert. Monitor nearby Tango Zulu Juliet on your right. Maintain visual separation.
		Pilot 1	Copy that, Air Traffic Control. We have a visual contact with Tango Zulu Juliet on our right.
		Pilot 2	Air Traffic Control, this is Tango Zulu Juliet. We have a TCAS resolution advisory. Climbing to avoid traffic.
		ATC	Tango Zulu Juliet, climb and maintain 5,000 feet. Continue to depart the area. I will inform the other aircraft.
		ATC	Alpha Bravo Charlie, this is Air Traffic Control. Traffic alert. Tango Zulu Juliet has a TCAS resolution advisory and is climbing to avoid traffic. Monitor your TCAS and maintain visual separation.

VII. TECHNICAL RESULTS

This section describes the evaluation results of our implemented text analytics anomaly detection system. First, we plotted the samples in our test data set within the learned VAE latent space. The 3D latent space is constructed from the principal component coefficients. Fig. 12 shows the latent space for the mean encodings characterized by the first three principal component metrics. Here, we can see that the normal test sample points have a relatively tight cluster of points shown in blue while the anomaly test sample points shown in yellow are further in distance from the normal cluster distribution. This shows that an effective latent space separating normal and anomalous test samples has been learned successfully.

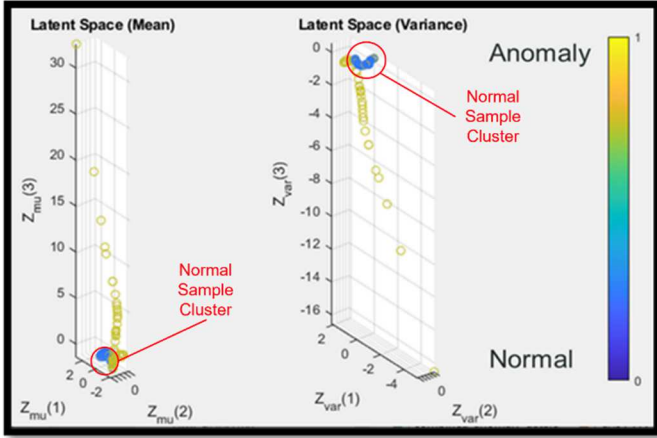


Fig. 12. Distribution of Normal and Anomalous Air Traffic Communication Samples Within the Learned VAE Latent Space

Furthermore, Fig. 13 shows the resulting test Receiver Operating Characteristic (ROC) curve. We can see that the AUC value is 0.8844 or 88.4%, which demonstrates that the majority of normal air traffic conversations are correctly classified as normal, and that the majority of anomalous air traffic conversations are correctly classified as anomalous.

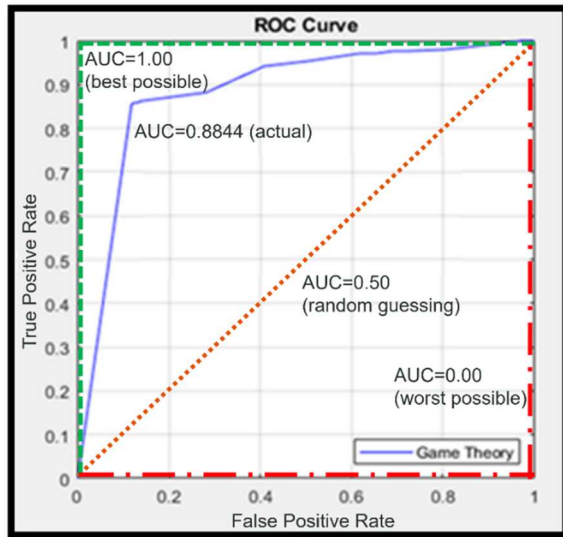


Fig. 13. Receiver Operating Characteristic (ROC) curve, with the area under curve ~88%

VIII. FUTURE WORK

For future work, we seek to continue implementing the modules in the Aviation Safety Anomaly Detection Framework. The Operational Regulatory Framework will need to be implemented and then factored into the anomaly detection pipeline through the use of fine-tuned LLMs. Additionally, the text analytics anomaly detection pipeline currently generates a nominal or off-nominal classification after processing the entirety of a conversational exchange, but in order for the system to be impactful on real-time ATC decision-making, a nominal/off-nominal judgment will need to be made after each individual turn in a conversation. Furthermore, LLMs combined with human factors considerations will be incorporated in order to generate an appropriate alert (e.g., auditory signal, visual cue, natural language response with recommended actions) to concisely convey the safety alerts and recommended actions to all involved parties without distracting from other essential stimuli. Finally, future work will incorporate Automatic Speech Recognition (ASR) to generate text transcriptions of ATC communications in real-time, similar to the work in [27].

IX. CONCLUSION

This work provides a 'Starting Point' that provides a benchmark for an 'off-nominal safety detection system' involving air-ground communication, ATC clearances, flight operations, etc. We employed Large Language Models to generate a synthetic dataset covering a diverse range of scenarios quickly and cheaply in both nominal and off-nominal aviation situations. We then used a combination of Variational Auto-Encoders and Text Analytics to differentiate between normal operations and anomalous situations posing safety-risks to the airspace.

Our experimental results indicated that a representation of vernacular used in normal flight operations could be generated to successfully identify communications containing verbiage outside of expected norms with high fidelity. Given these promising results, we anticipate that further use of LLMs and generative AI will facilitate the enablement of aviation safety systems to communicate identified safety risks back to cognizant ATCs and flight crews, thus enhancing situational awareness of all parties in a role appropriate manner.

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